

EXPERIMENT 6

SELECT with Various Clauses – BETWEEN, IN, Aggregate Function

AIM:

To view the records from the tables using SELECT commands with BETWEEN, IN, Aggregate functions.

DESCRIPTION:

- The BETWEEN operator allows you to specify a range to test. We often use the BETWEEN operator in the WHERE clause of the SELECT, INSERT, UPDATE, and DELETE statements.
- The IN operator allows you to determine if a specified value matches any one of a list or a sub query.
- MySQL provides many aggregate functions that include AVG, COUNT, SUM, MIN, MAX, etc. An aggregate function ignores NULL values when it performs calculation except for the COUNT function.

QUESTIONS

IN & BETWEEN

1. List the details of patients whose gender is either '**M**' or '**F**' using the IN operator.
2. List the patients whose age is in the range **20 to 30** using the BETWEEN operator.

AGGREGATE FUNCTIONS

3. Find the **average age** of all patients.
4. Find the **maximum and minimum age** of patients.
5. Find the **maximum, minimum and average age** of patients for each gender.
6. Display the **gender and average age** of patients for each gender.

7. Calculate the **sum of the ages** of all patients.

8. Find how many patients are registered for each **gender**. Display the gender and number of patients.

9. Find how many appointments are assigned to each **doctor**. Display the Doctor ID and number of appointments.

OUTPUT:

1)

```
39 • SELECT *
```

```
30 FROM Patient
```

```
31 WHERE Gender IN ('M','F');
```

```
32
```

sult Grid |  Filter Rows: | Edit:    | Export/Import

Patient_ID	Patient_Name	Age	Gender	Disease	Phone_No
P001	Arun Kumar	25	M	Fever	9876543210
P002	Priya	22	F	Diabetes	9876543211
P003	Rahul	30	M	Asthma	9876543212
P004	Sneha	28	F	Migraine	9876543213
NULL	NULL	NULL	NULL	NULL	NULL

2)

```
194 • SELECT *
```

```
195 FROM Patient
```

```
196 WHERE Age BETWEEN 20 AND 30;
```

```
197
```

Result Grid |  Filter Rows: | Edit:    | Export/Import: 

	Patient_ID	Patient_Name	Age	Gender	Disease	Phone_No
▶	P001	Arun Kumar	25	M	Fever	9876543210
	P002	Priya	22	F	Diabetes	9876543211
	P003	Rahul	30	M	Asthma	9876543212
	P004	Sneha	28	F	Migraine	9876543213
*	NULL	NULL	NULL	NULL	NULL	NULL

3)

```
199 • SELECT AVG(Age) AS Average_Age
200 FROM Patient;
201
```

Result Grid			Filter Rows:		Export:		Wrap Cell Cont
	Average_Age						
▶	26.2500						

4)

```
203 • SELECT MAX(Age) AS Maximum_Age,
204 MIN(Age) AS Minimum_Age
205 FROM Patient;
206
```

Result Grid			Filter Rows:		Export:		Wrap C
	Maximum_Age	Minimum_Age					
▶	30	22					

5)

```
208 • SELECT Gender,
209 MAX(Age) AS Maximum_Age,
210 MIN(Age) AS Minimum_Age,
211 AVG(Age) AS Average_Age
212 FROM Patient
213 GROUP BY Gender;
214
```

Result Grid			Filter Rows:		Export:		Wrap Cell C
	Gender	Maximum_Age	Minimum_Age	Average_Age			
	M	30	25	27.5000			
	F	28	22	25.0000			

6)

```
216 • SELECT Gender,  
217         AVG(Age) AS Average_Age  
218 FROM Patient  
219 GROUP BY Gender;  
220
```

Result Grid   Filter Rows: Export:  Wrap C		
	Gender	Average_Age
▶	M	27.5000
	F	25.0000

7)

```
22 • SELECT SUM(Age) AS Total_Age  
23 FROM Patient;  
24
```

Result Grid   Filter Rows: Export	
	Total_Age
	105

8)

```
226 • SELECT Gender,  
227         COUNT(*) AS Number_of_Patients  
228 FROM Patient  
229 GROUP BY Gender;  
230
```

Result Grid   Filter Rows: Export:  Wrap Cell Co		
	Gender	Number_of_Patients
▶	M	2
	F	2

9)

```
232 • SELECT Doctor_ID,  
233         COUNT(*) AS Number_of_Appointments  
234 FROM Appointment  
235 GROUP BY Doctor_ID;  
236
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Doctor_ID	Number_of_Appointments
·	D001	1
	D002	1
	D003	1
	D004	1

RESULT:

The records from the tables are displayed using SELECT commands with WHERE Clause and Pattern matching.

